

UNIT 7&8 Part 4:

Elasticity of Demand



Why do the demand for different products react differently to price changes?

Elastic Demand



Inelastic Demand



UNIT 7&8 Part 4 Overview

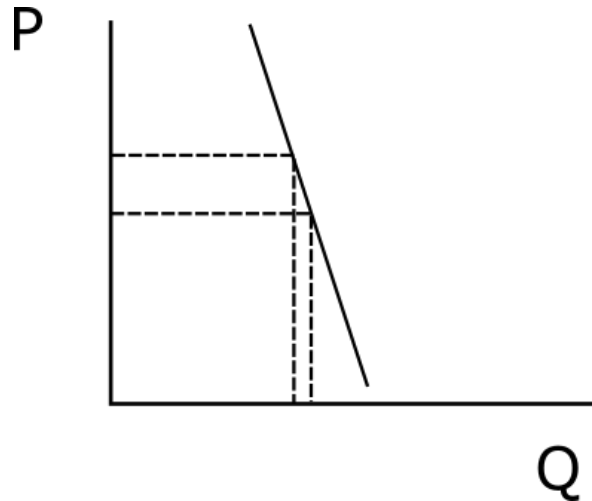
- Concepts
- Elasticity Formula
- Elasticity and marginal revenue
- Elasticity and market power

7.5 Price elasticity of demand

- How sensitive is the quantity demanded (Q_d) for a certain product to a change in that product's price?
 - Firms want to know how their sales quantity will react to price changes.
 - Governments want to know how markets will react to taxes.
- **Price-elastic** demand has a **large** Q_d response to a given price change.
- **Price-inelastic** demand has a **small** Q_d response to a given price change.

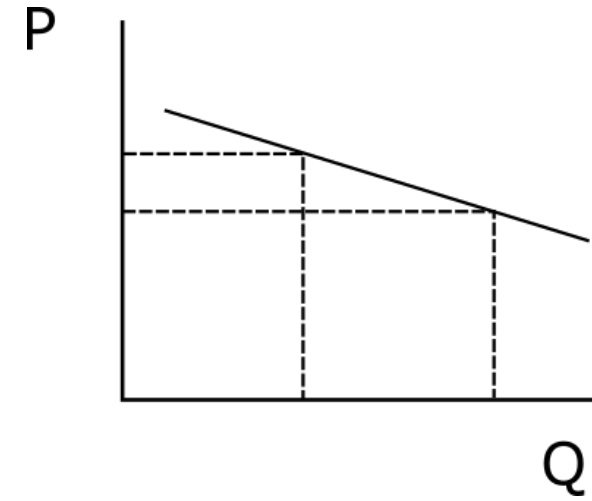
7.5 Price elasticity of demand

The **slope** of a demand curve measures (inversely) quantity response to price change:



Relatively steep D curve

Q_d relatively **insensitive** to change in P
(P relatively sensitive to change in Q)



Relatively flat D curve

Q_d relatively **sensitive** to change in P
(P relatively insensitive to change in Q)

But the slope depends on the units of measurement, e.g. measuring quantities in kilograms instead of tonnes will make the slope 1000 times flatter.

7.5 Price elasticity of demand

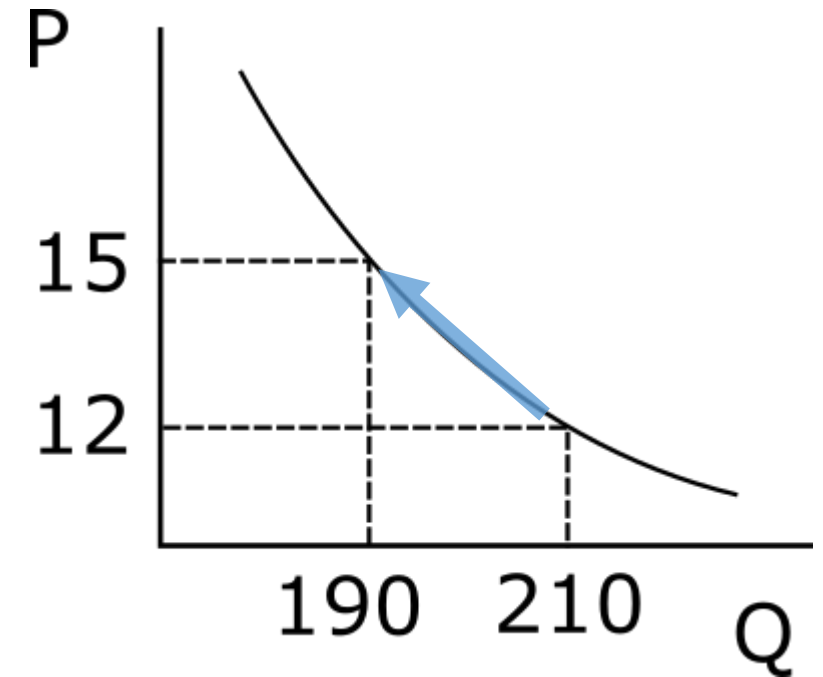
A better (unit-free) measure uses the ratio of percentage changes:

- With what percentage will quantity demanded change given a 1 per cent increase in price?
- $\epsilon = (-) \frac{\% \text{ change in quantity demand}}{\% \text{ change in price}} = (-) \frac{\% \Delta Q_d}{\% \Delta P}$
- Work with *percentage* changes and NOT change in absolute numbers
- Minus sign: We do not include the minus sign, intuitively the formula would have given a negative answer as suggested above (because Q_d and P move in opposite directions). However, by convention economists refer to the number as a positive number, e.g. if you do not include the minus sign in the formula and get an answer of -4 , you would still say, “the price elasticity of demand is four”; you would also say “the elasticity of demand is larger than one” (despite $-4 < -1$). For the purposes of this course we use an absolute value.
- Percentage change calculation: “multiplying by 100” optional

Price elasticity of demand

E.g., say price increases from 12 to 15 while quantity demanded decreases from 210 to 190.

- $\% \Delta Q_d = \frac{190-210}{210} \times 100 = -9.5$ (or -0.095)
- $\% \Delta P = \frac{15-12}{12} \times 100 = 25$ (or 0.25)
- $\epsilon = (-) \frac{\% \Delta Q_d}{\% \Delta P} = (-) \frac{-9.5}{25} = (-) \frac{-0.095}{0.25}$
 $= (-) - 0.38 = 0.38$
- Because $\epsilon < 1$, this demand curve is **inelastic**.

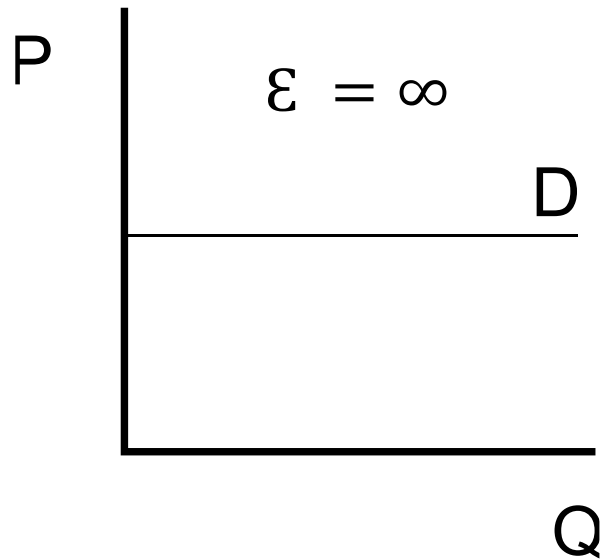


$\epsilon > 1$: Elastic
 $\epsilon < 1$: Inelastic
 $\epsilon = 1$: Unit elastic

Extreme cases

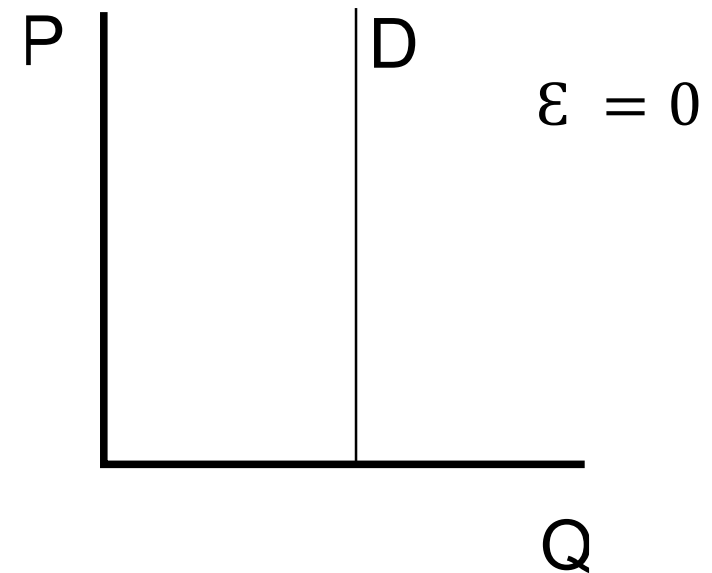
Perfectly price-elastic demand

- **Any** quantity can be sold at a specific price
- Demand for a firm's product in a perfectly competitive market



Perfectly price-inelastic demand

- Regardless of the price, a specific quantity will be demanded



Price Elasticity of Demand

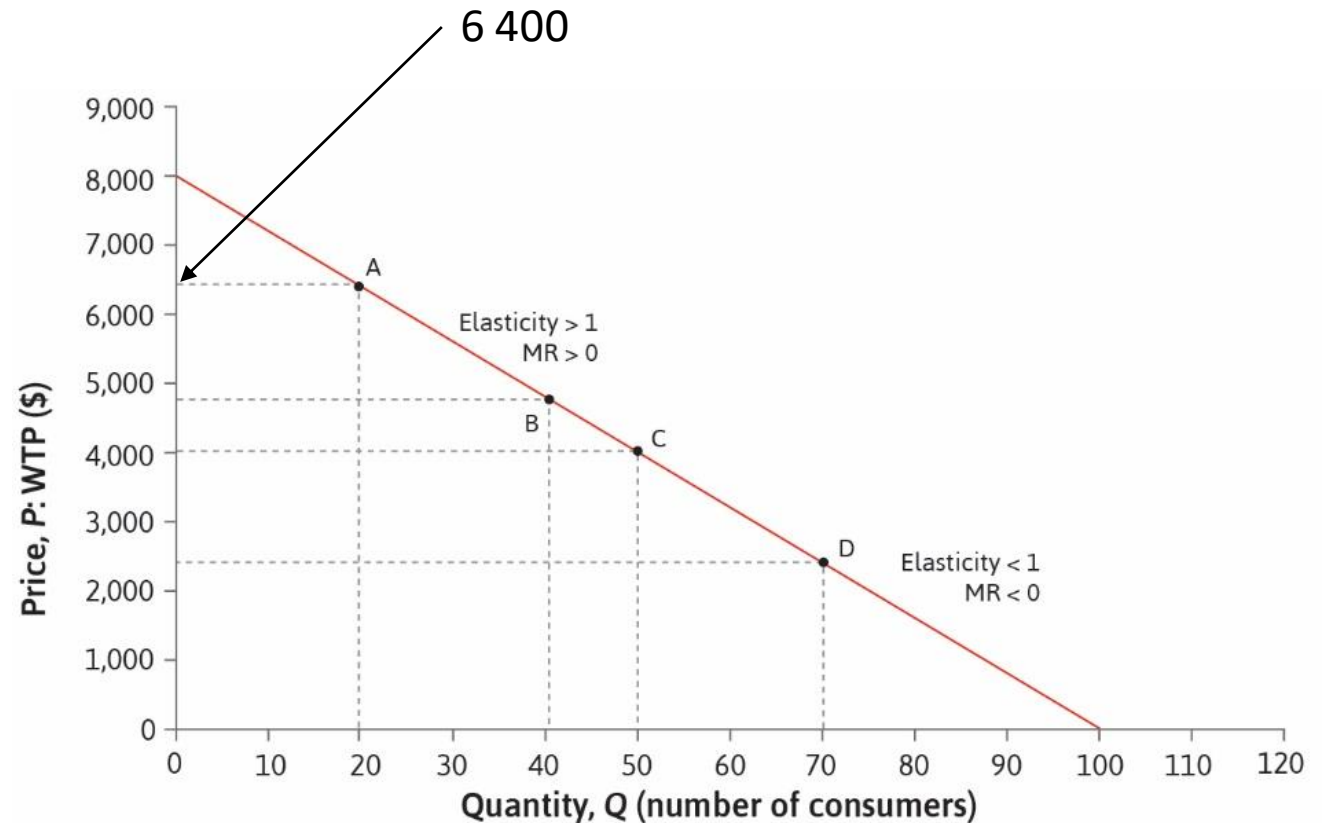
Slope of demand curve = -80
 Slope the same at any point on
 linear (straight line) demand curve,
 so $\Delta Q_d = 1$ and $\Delta P = -80$.

At A:

$$\% \Delta Q_d = \left(\frac{\Delta Q}{Q} \right) \times 100 = \left(\frac{1}{20} \right) \times 100 = 5$$

$$\% \Delta P = \left(\frac{\Delta P}{P} \right) \times 100 = \left(\frac{-80}{6400} \right) \times 100 = -1.25$$

$$\epsilon = - \frac{5}{-1.25} = 4$$



Price Elasticity of Demand

At **A**:

$$\text{If } Q = 19: P = 8\,000 - 80(19) = 6\,480$$

$$TR_{Q=19} = 6\,480 \times 19 = 123\,120$$

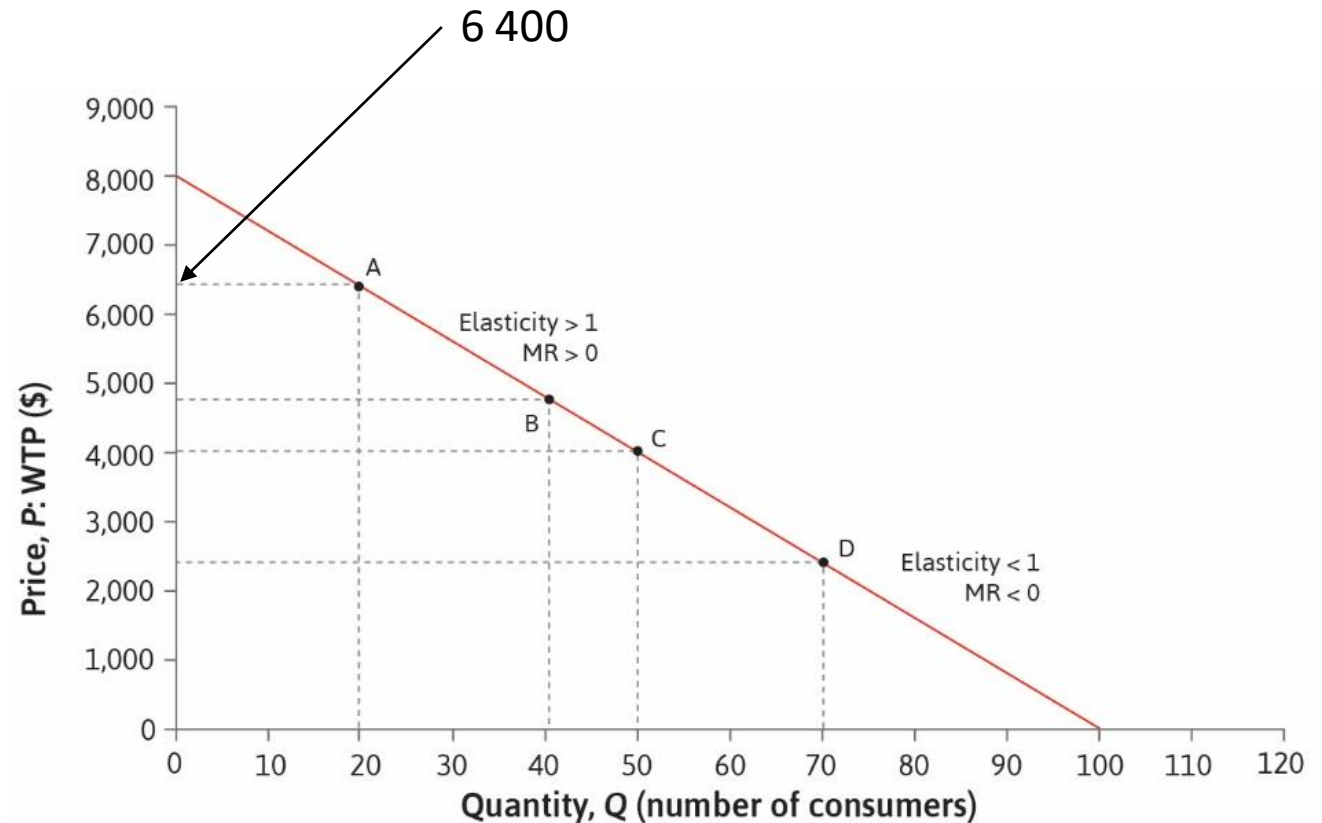
$$\text{If } Q = 20: P = 8\,000 - 80(20) = 6\,400$$

$$TR_{Q=20} = 6\,400 \times 20 = 128\,000$$

$$\text{Thus } MR_{Q=20} = 128\,000 - 123\,120 = 4\,880$$

When $\varepsilon > 1$, $|\% \Delta Q_d| > |\% \Delta P|$, thus

$MR > 0$ when $Q_d \uparrow$ and $P \downarrow$.



Price Elasticity of Demand

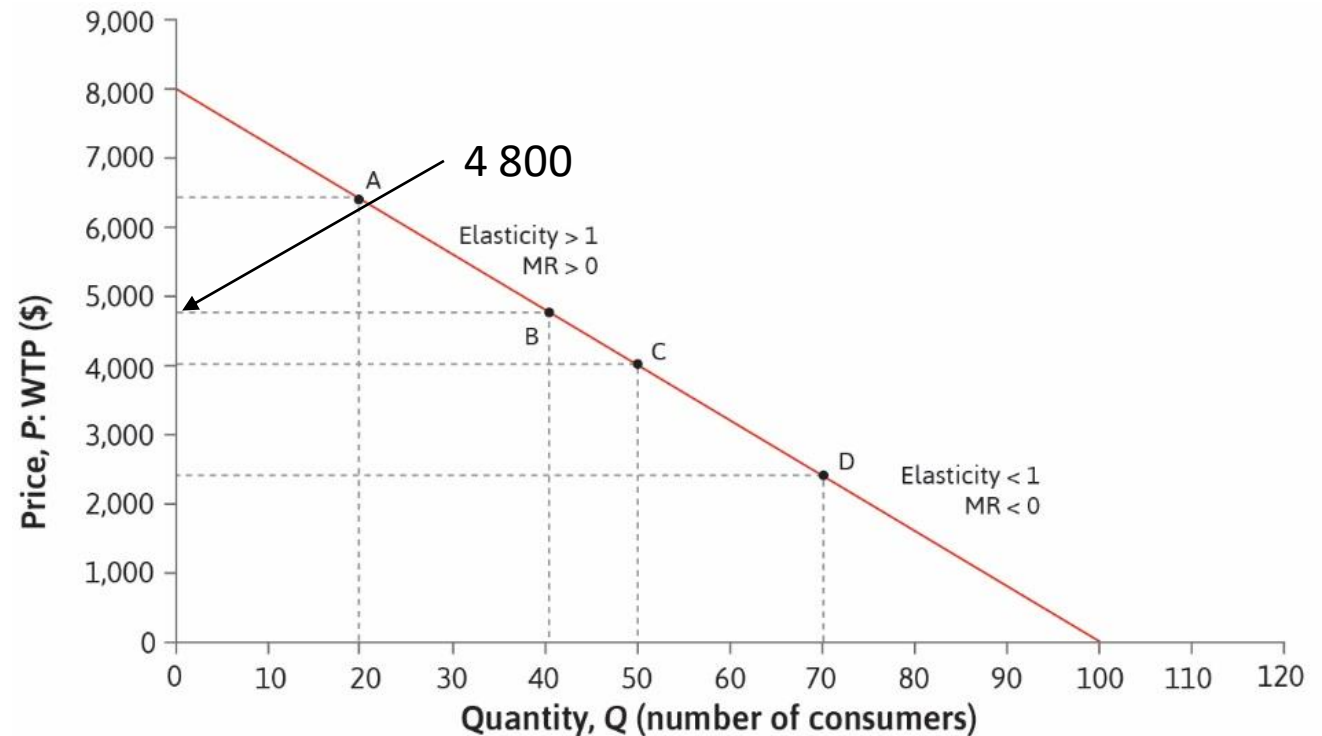
Slope of demand curve = -80

At **B**:

$$\% \Delta Q_d = \left(\frac{\Delta Q}{Q} \right) \times 100 = \left(\frac{1}{40} \right) \times 100 = 2.5$$

$$\% \Delta P = \left(\frac{\Delta P}{P} \right) \times 100 = \left(\frac{-80}{4800} \right) \times 100 = -1.67$$

$$\varepsilon = - \frac{2.5}{-1.67} = 1.5$$



Price Elasticity of Demand

At **B**:

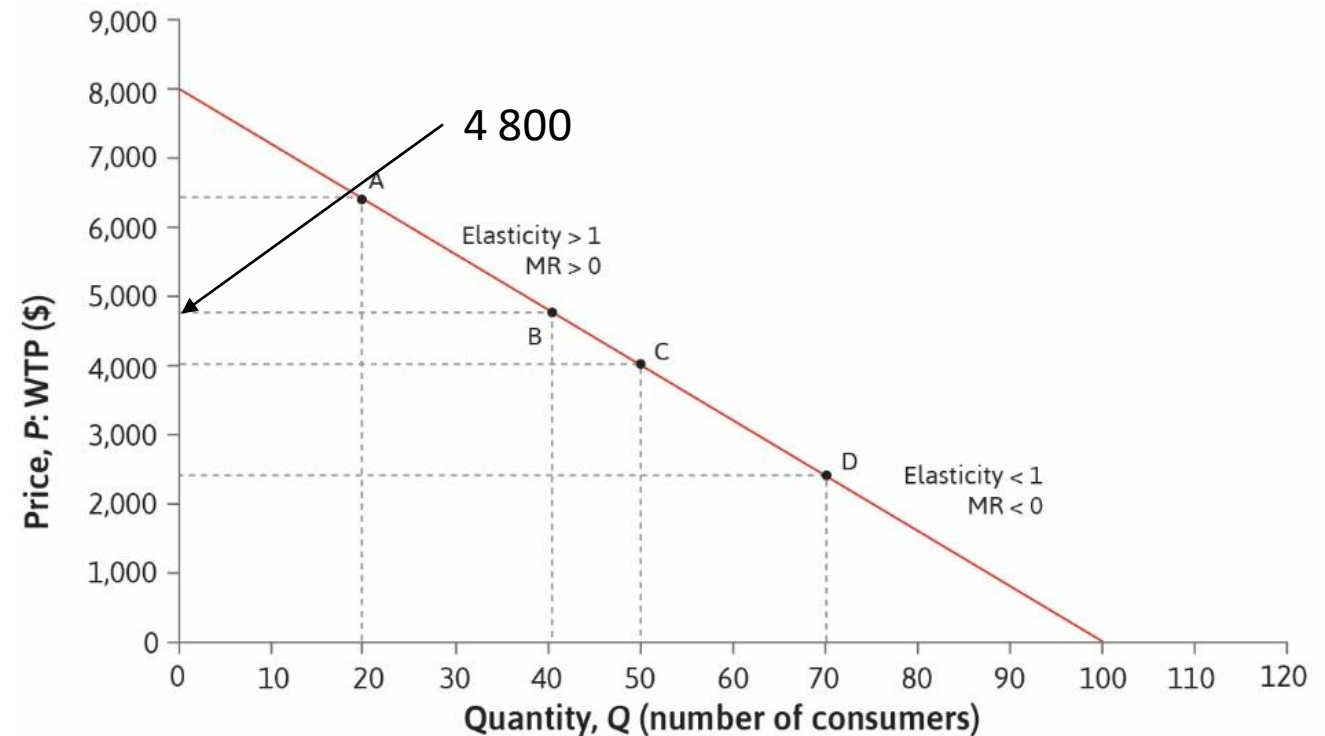
$$\text{If } Q = 39: P = 8\,000 - 80(39) = 4\,880$$

$$TR_{Q=39} = 4\,880 \times 39 = 190\,320$$

$$\text{If } Q = 40: P = 8\,000 - 80(40) = 4\,800$$

$$TR_{Q=40} = 4\,800 \times 40 = 192\,000$$

$$\text{Thus } MR_{Q=20} = 192\,000 - 190\,320 = 1\,680$$



Price Elasticity of Demand

On a linear demand curve:

Top half of demand curve

$$\varepsilon > 1$$

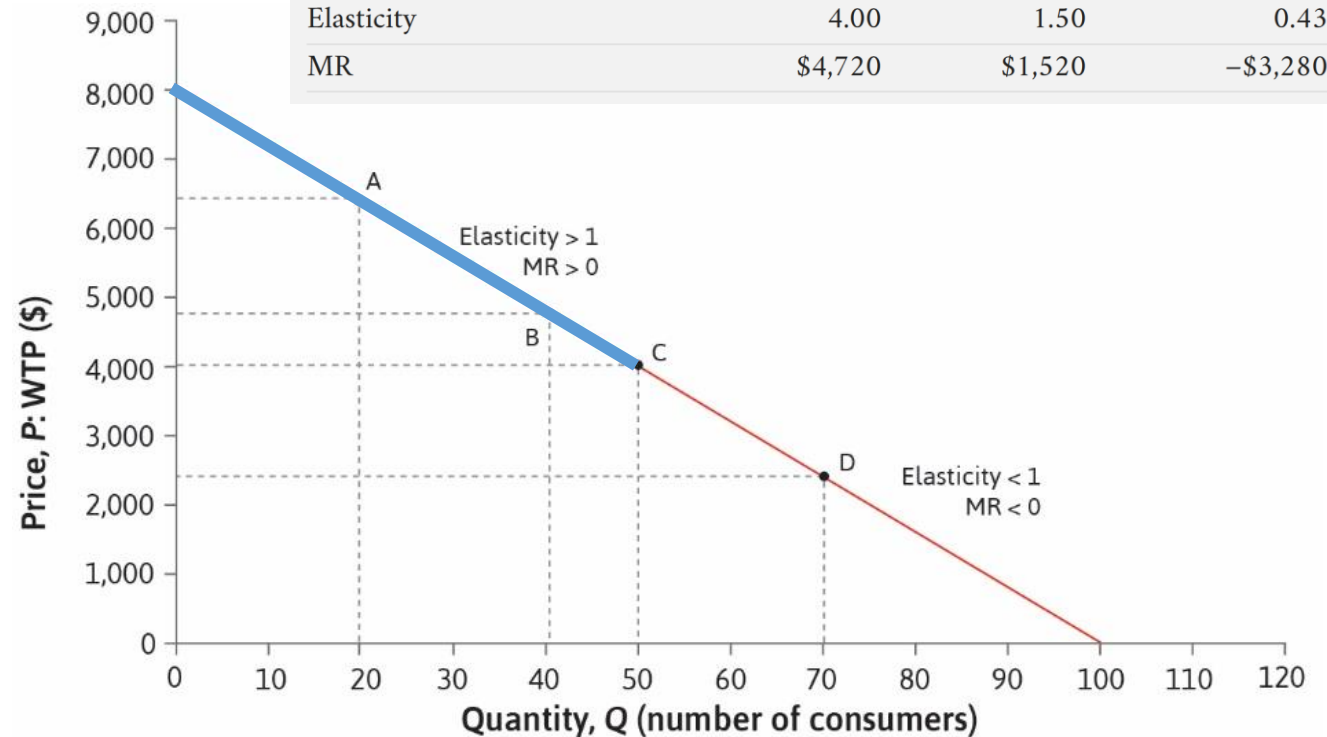
$$|\% \Delta Q_d| > |\% \Delta P|$$

MR positive

As P decreases TR increases

As P increases TR decreases

Elasticity = - % Change in Q / % Change in P			
	A	B	D
Q	20	40	70
P	\$6,400	\$4,800	\$2,400
ΔQ	1	1	1
ΔP	-\$80	-\$80	-\$80
% change in Q	5.00	2.50	1.43
% change in P	-1.25	-1.67	-3.33
Elasticity	4.00	1.50	0.43
MR	\$4,720	\$1,520	-\$3,280



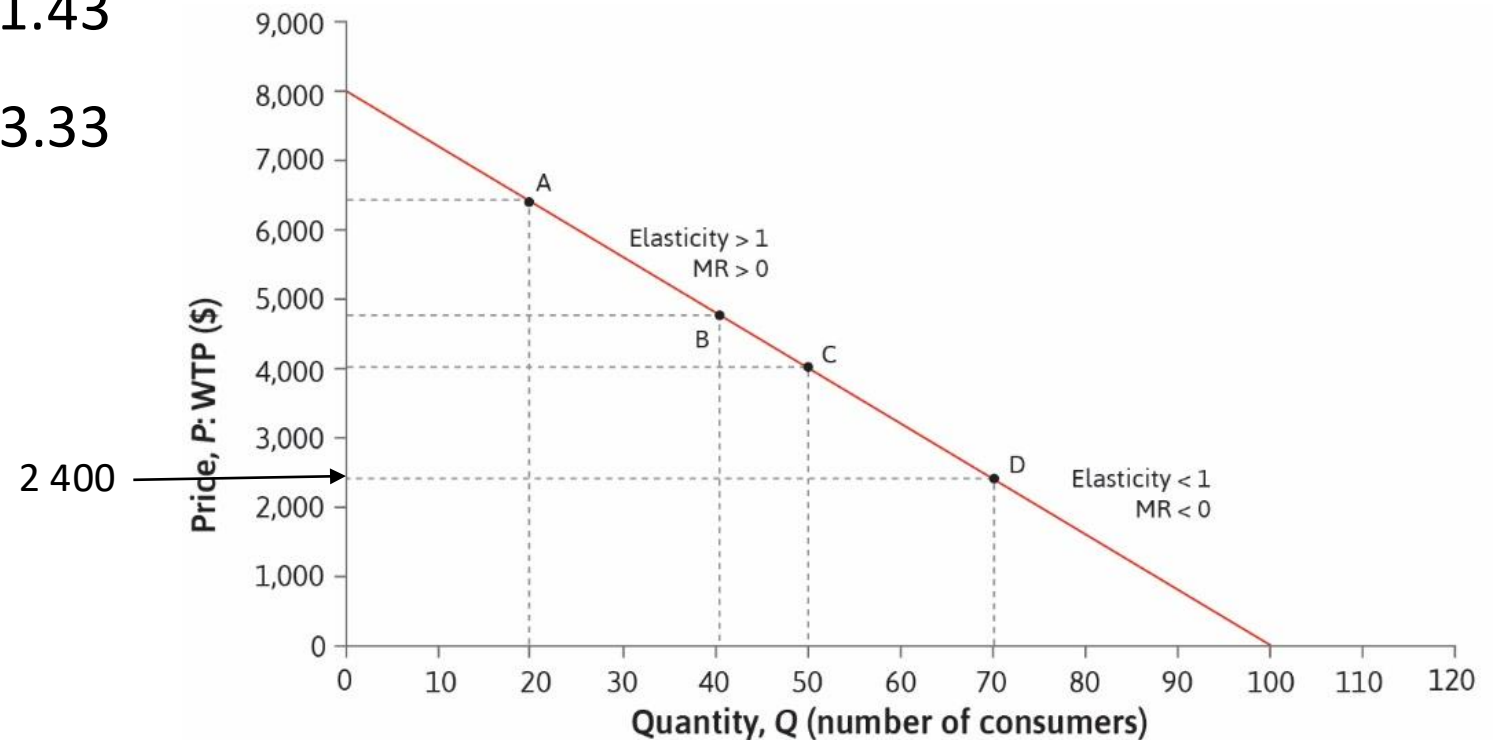
Price Elasticity of Demand

At **D**:

$$\% \Delta Q_d = \left(\frac{\Delta Q}{Q} \right) \times 100 = \left(\frac{1}{70} \right) \times 100 = 1.43$$

$$\% \Delta P = \left(\frac{\Delta P}{P} \right) \times 100 = \left(\frac{-80}{2,400} \right) \times 100 = -3.33$$

$$\epsilon = - \frac{1.43}{-3.33} = 0.43$$



Price Elasticity of Demand

At **D**:

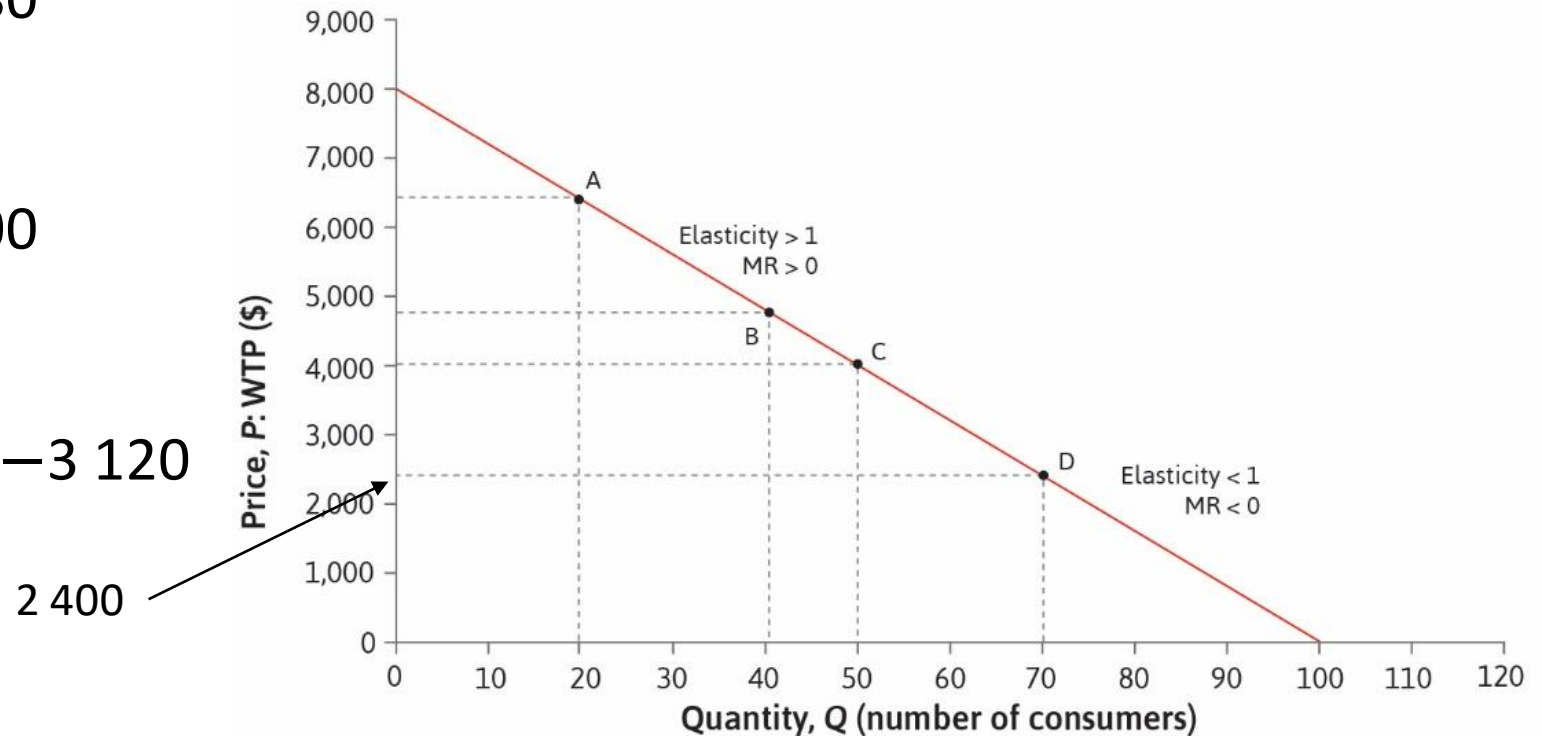
$$\text{If } Q = 69: P = 8\,000 - 80(69) = 2\,480$$

$$TR_{Q=69} = 2\,480 \times 69 = 171\,120$$

$$\text{If } Q = 70: P = 8\,000 - 80(70) = 2\,400$$

$$TR_{Q=70} = 2\,400 \times 70 = 168\,000$$

$$\text{Thus } MR_{Q=20} = 168\,000 - 171\,120 = -3\,120$$



When $\epsilon < 1$, $|\% \Delta Q_d| < |\% \Delta P|$, thus
 $MR < 0$ when $Q_d \uparrow$ and $P \downarrow$.

Price Elasticity of Demand

On a linear demand curve:

Bottom half of demand curve

$$\varepsilon < 1$$

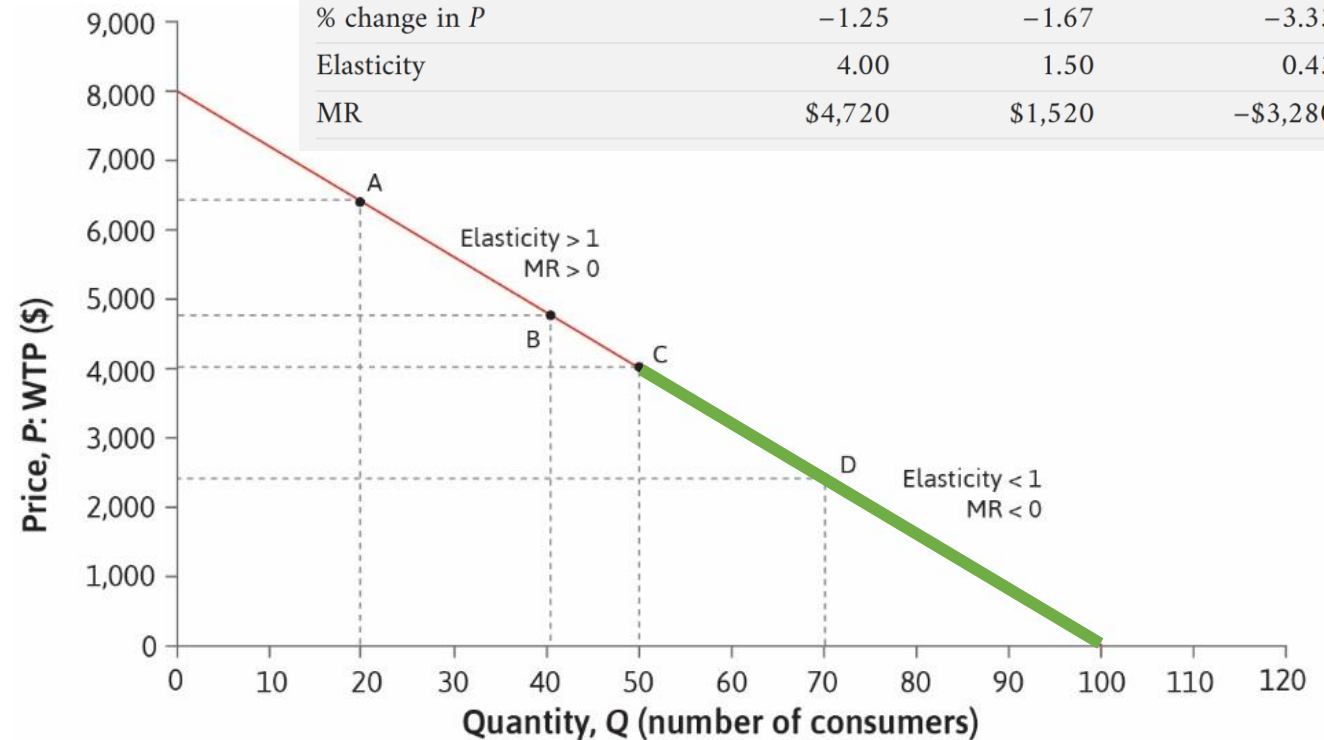
$$|\% \Delta Q_d| < |\% \Delta P|$$

MR negative

As P decreases TR decreases

As P increases TR increases

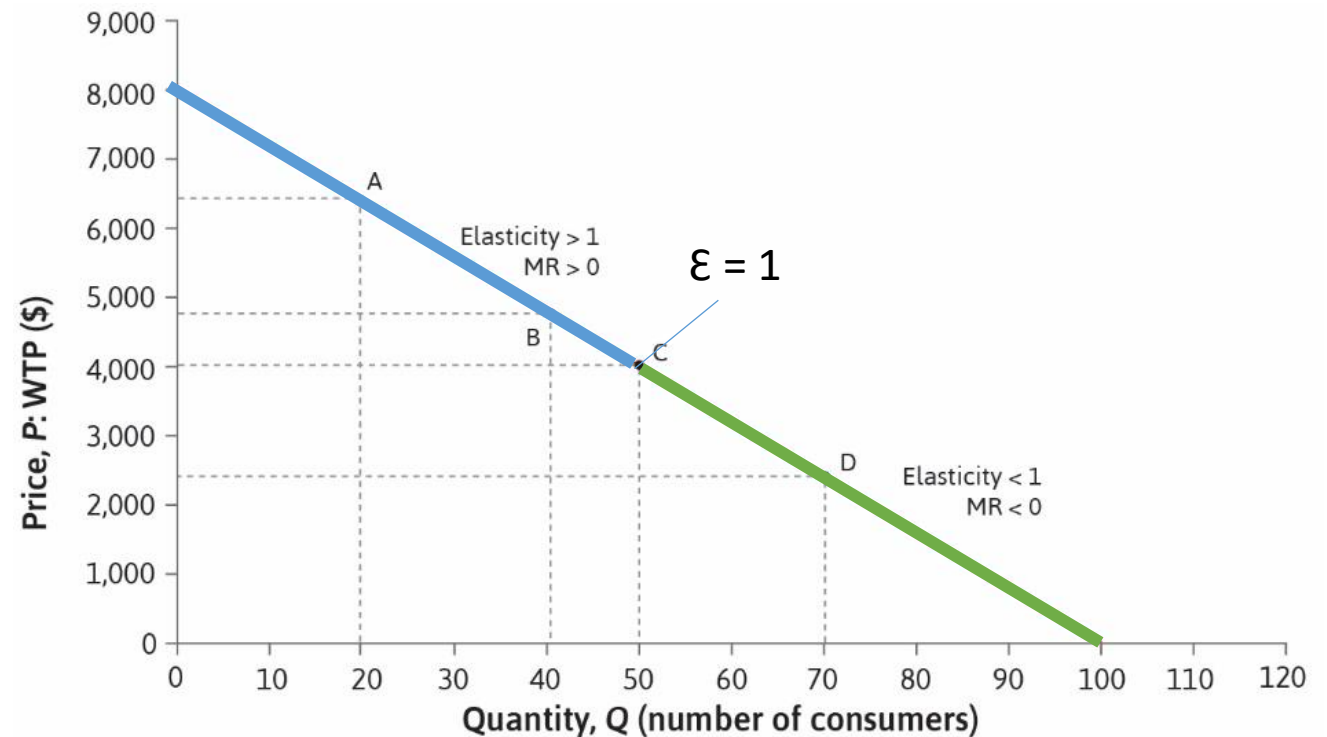
Elasticity = - % Change in Q / % Change in P			
	A	B	D
Q	20	40	70
P	\$6,400	\$4,800	\$2,400
ΔQ	1	1	1
ΔP	-\$80	-\$80	-\$80
% change in Q	5.00	2.50	1.43
% change in P	-1.25	-1.67	-3.33
Elasticity	4.00	1.50	0.43
MR	\$4,720	\$1,520	-\$3,280



Price Elasticity of Demand

At **C**:

- $|\% \Delta Q_d| = |\% \Delta P|$
- $\epsilon = 1$
- $MR = 0$
- TR is maximized



Exercise

- Take the given demand curve: $P = 8\,000 - 80Q_d$
- Draw a graph of the TR – with TR on the vertical axis and quantity on the horizontal axis.
- Calculate the TR curve when Q_d is: 0; 20; 40; 50; 70; and 100.
- How can you use this graph to guide you with your pricing policy?

Price Elasticity of Demand

The factors that influence the price elasticity of demand:

- The availability of close substitutes
- How broadly defined the product category is
- Whether the product is essential or optional
- The proportion of income spent on the good
- The time elapsed since a price change

Price Elasticities of Demand

Good or Service	Elasticity
Elastic Demand	
Metals	1.52
Electrical engineering products	1.39
Mechanical engineering products	1.30
Furniture	1.26
Motor vehicles	1.14
Instrument engineering products	1.10
Professional services	1.09
Transportation services	1.03
Inelastic Demand	
Gas, electricity, and water	0.92
Chemicals	0.89
Drinks (all types)	0.78
Clothing	0.64
Tobacco	0.61
Banking and insurance services	0.56
Housing services	0.55
Agricultural and fish products	0.42
Books, magazines, and newspapers	0.34
Food	0.12
Oil	0.05

Critical Awareness: Limitations of Elasticities



Problems with
inaccurate or
incomplete data
collection



Consumer price sensitivity changes over time



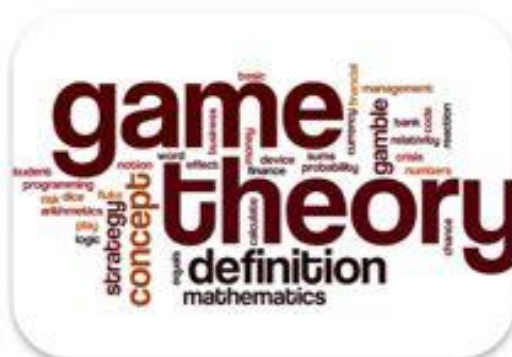
Elasticity of demand varies by region / time



Not all businesses are profit maximisers



Elasticity will vary within product ranges
e.g. economy and premium products

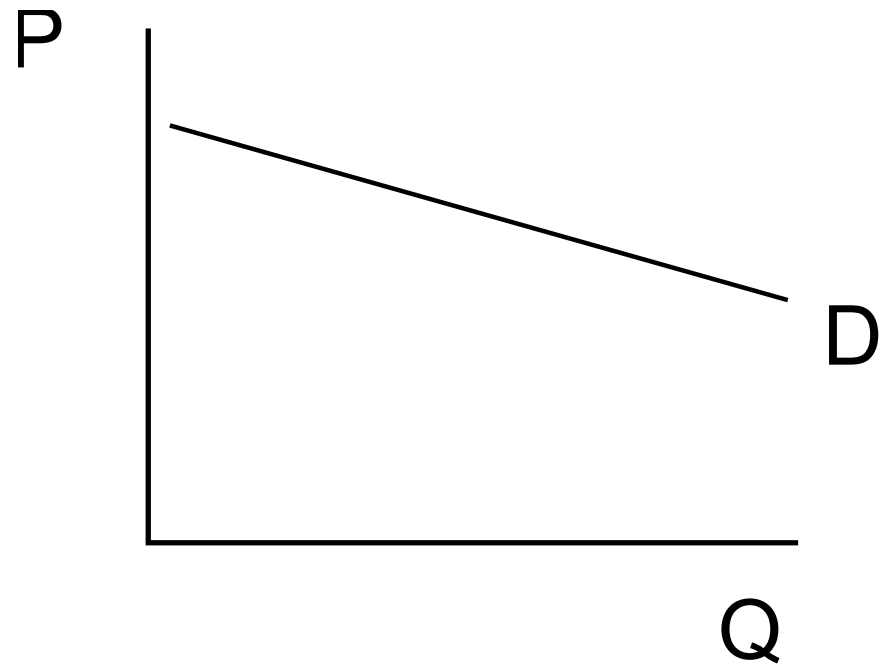


Rival producers will change their market strategies from time to time

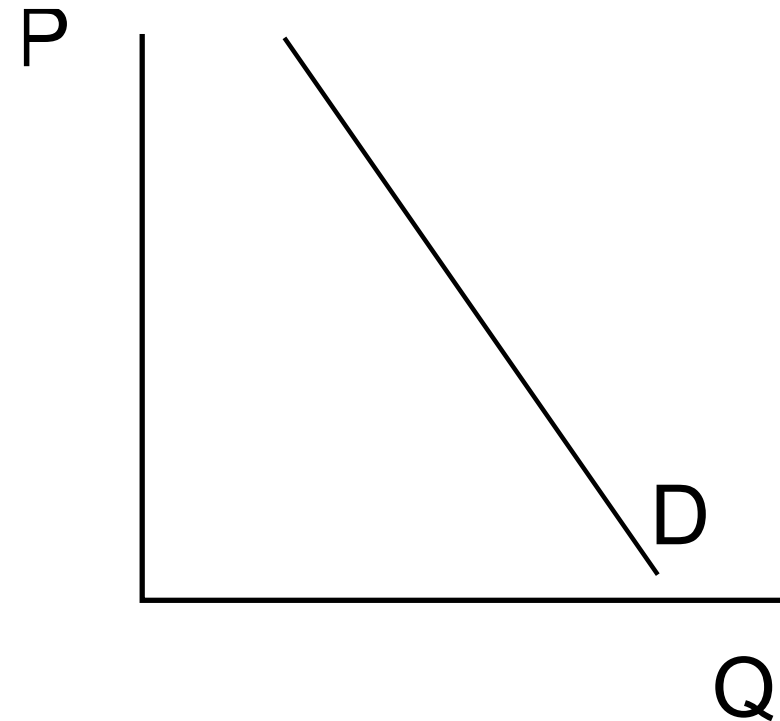
Price Elasticity and Profits

If you draw demand curves on the same scale:

Products with more elastic demand generally have flatter curves.



Elastic demand

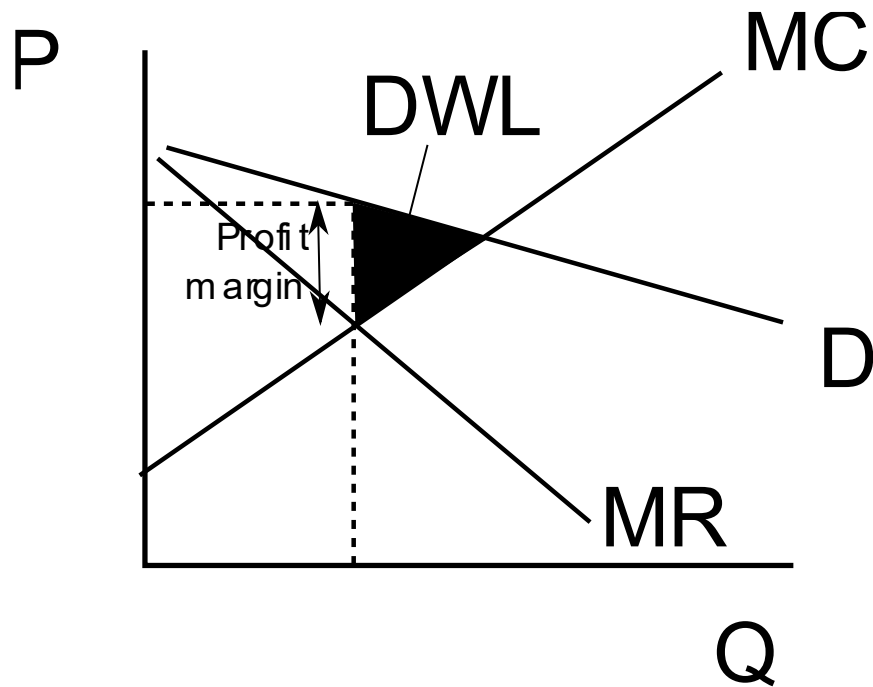


Inelastic demand

Price Elasticity and Profits

With an elastic demand a firm's **profit margin** is small
Do not have much market power to increase prices.

Deadweight loss is small.

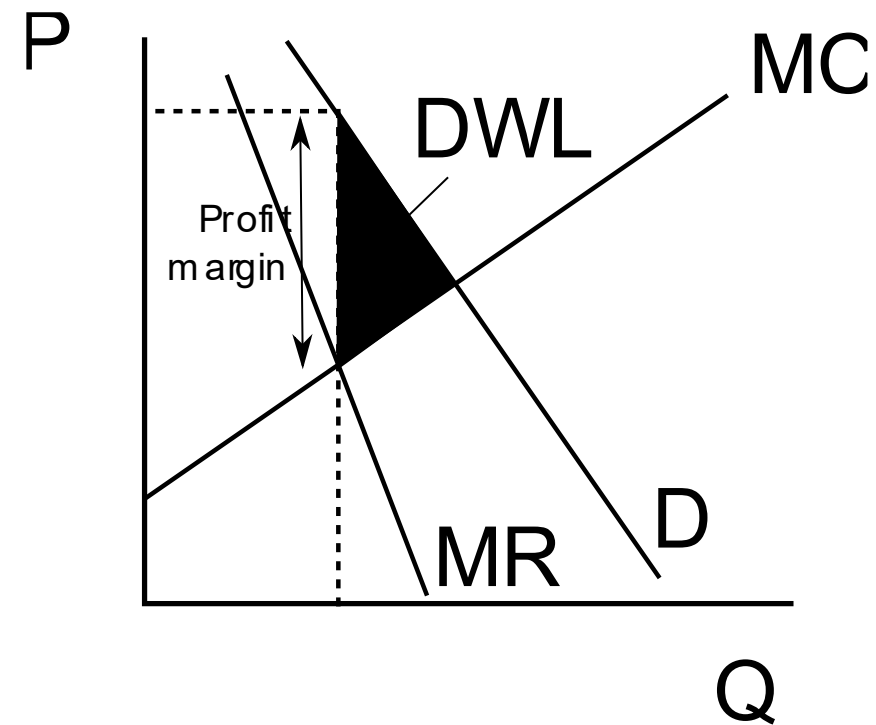


Elastic demand



Price Elasticity and Profits

With an inelastic demand a firm's **profit margin** is large
Do have market power to increase prices.
Deadweight loss is large.



Inelastic demand

Summary: Price Elasticity

An *elastic demand* is very responsive to price changes

If you raise prices by 10%, quantity demanded will decrease by more than 10% and TR will decrease.

An *inelastic demand* is not very responsive to price changes

If you raise prices by 10%, quantity demanded will decrease by less than 10% and TR will increase.

If the demand for your product is inelastic you can charge a higher **profit margin** (thus typically higher economic profits) and vice versa.

Exercise 1

1. If P increases and ε is smaller than 1, what happens to TR ?
2. If Q increases and ε is smaller than 1, what happens to TR ?
3. If Q increases and TR increases, what is ε ?
4. If P decreases and TR decreases, what is ε ?

Remember:

$$TR = P \times Q$$

$$\varepsilon = (-)(\% \Delta Q / \% \Delta P)$$

Exercise 2

If your firm asks R60 per hamburger the quantity demanded is 1 500.

You lower the price to R48 and now you sell 2 400 hamburgers.

- a) Calculate the price elasticity of demand for burgers if the price decreases.
- b) Given your answer, what price policy must you follow to increase your income? Explain your answer.